Initial Results From "Beyond Evaluation Use": A Study of Involvement and Influence in Large, Multi-site National Science Foundation (NSF) Evaluations





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Background: The "Beyond Evaluation Use" Grant

- 3-year grant to study use, influence, and involvement within NSF-funded programs.
- Research team (2 co-PIs and 4 graduate students) based at the University of Minnesota.

TARGETED CASE STUDIES

FOCUS QUESTIONS:

(1) What is the relationship between involvement in the multi-site evaluation and the use and influence of the evaluation on project leaders and evaluators?

(2) What factors appear to be related to the use and influence of the evaluation on the projects?

CONCEPTUAL FRAMES

- Evaluation use/ utilization and influence literature
- Participatory evaluation literature

METHODS

- Online survey (project leaders and evaluators)
- Interviews (project leaders and evaluators; NSF program officers)
- Review of archival documents, publications, and reports produced

INVOLVEMENT OF PROJECT LEADERS AND EVALUATORS, THEIR USE OF 4 NSF PROGRAM EVALUATIONS, AND THE EVALUATIONS' INFLUENCE

Four Cases

- Advanced Technological Education (ATE)
  (Evaluated: 1998-2005)
- Building Evaluation Capacity of STEM Projects - Math Science Partnership (MSP-RETA)
  (Evaluated: 2002 – 2005)
- Collaboratives for Excellence in Teacher Preparation (CETP) (Evaluated: 1999 – 2005)
- Local Systemic Change through Teacher Enhancement (LSC) (Evaluated: 1995 – 2006)

The Pros and Cons of Annual Surveys: A Case Study of the ATE Program Evaluation

- The online survey seemed to overshadow the other evaluation activities.
- A low feeling of involvement other than completing the survey.
- Acknowledgment of the importance of the evaluation as a necessary step in securing programmatic funding.
- The lack of human/financial resources and the generality of the survey limited the impact of the evaluation.
- Appreciation of the collaboration fostered by the evaluation.
- Increased awareness about meeting the project goals.

The Importance of Planning: Involvement, Use, and Influence in the CETP Core Evaluation

- Levels and types of involvement in the CETP core evaluation varied across projects and individuals.
- Primary involvement in the planning/instrument development and the data gathering phases.
- The majority of respondents were involved at least a little or some in evaluation.
- The greatest influence was on project leaders' and evaluators' knowledge and understandings, followed by influences on their skills, beliefs, and use in other contexts.
- Involvement in planning meetings fostered commitment to the evaluation process and confidence in the evaluation instruments.
- The late start of the core evaluation minimized its influence.

Does Local Ownership Improve National Evaluation?

A Case Study of the LSC Core Evaluation

- Evidence of the impact of the core evaluation on project leaders and evaluators' knowledge, skills, and beliefs.
- Involvement in the evaluation activities is associated with the use of the evaluation in later projects and activities.
- Higher levels of involvement in the LSC core evaluation by the evaluators – compared with the project leaders.
- More people reported having volunteered to be involved than required, despite the mandatory participation in the LSC core evaluation.

Tracking the Influence of Building Evaluation Capacity in STEM Education: A Case Study of the Utah State MSP-RETA

- Evaluation capacity building rather than program evaluation.
- Two distinct groups of participants: people interacting with hands-on TA experts and attending meetings.
- The main lasting effect on the STEM field was the better understanding of evaluation and stronger community building.
- Enabling factors: responsiveness/expertise of the MSP-RETA leadership; participation in conferences; rigorous evaluation models/analysis; writing/presenting evaluation reports.
- Limiting factors: other competing priorities; lack of responsiveness and the non-applicability of the MSP-RETA's framework in some cases.

Cross-case Findings/Themes

- The importance of lifecycles
  - Program-evaluation level (e.g., timing of funding in LSC and CETP, input from different groups at different times in ATE)
  - Project-evaluation level (e.g., different experiences by “mature” and “developing” CETP projects, significant evolution of the MSPs and the RETA)
  - Individual level (e.g., previous evaluation experience, retirements, moves)

Cross-case Findings/Themes (Cont.)

- Project control
- Project uniqueness (i.e., alignment between the program and project evaluations)
- Other
  - Networking and dissemination in all 4 cases
  - Responsiveness and expertise of the MSP-RETA leadership
  - A variety of roles of data collection in LSC

Additional Research Plans

- Citation analysis as an indicator of spread of the ideas to the wider community
- Examination of the validity of a new evaluation use scale
- In-depth investigation of high users of the program evaluations
- Further study of the historical and cultural context of the program evaluations
- Examination of feasibility of applying network analysis
- A volume of *New Directions for Evaluation*

ATE Evaluation Logic Model

[1/25/2007]

Evaluation Inputs

NSF Grant Funding
- $1.3 million (1999-2002)
- $1.8 million (2003-2006)

NSF Goals for ATE Evaluation
- Provide evidence of quality of ATE program
- Inform program improvement

Evaluation Questions
- To what degree is the program achieving its goals?
- Is it making an impact and reaching the individuals and groups intended?
- How effective is it when it reaches its constituents?
- Are there ways to significantly improve the program?

ATE Program Components
- Grantee characteristics
- Organizational Practices
- Collaboration
- Materials Development
- Program Improvement
- Professional Development
- Recruitment/Retention

Personnel/Expertise
- NSF 3-person mgmt team
- Experienced evaluation team
- Eval Advisory Committee
- Evaluation Advisory Committee
- ATE centers and projects
- Local evaluators and info
- Program participants – institutions, faculty, students
- Content experts

Evaluation Activities

Data Collection
- Surveys: Annual, web-based
- Site Visits: Identify sites' major activities and accomplishments

Validate Survey Findings
- Sites Visits: 13 sites

Data Analysis
- Commission 9 issue papers to synthesize site visit reports, survey data, and existing research
- Synthesize findings with literature
- Develop 4 targeted studies

Meetings
- NSF annual PI meetings
- Advisory Committee Meetings
- Evaluation Adv Committee

Metaevaluation
- Hartmann, Survey System Eval
- Stake, Site Visits & Issue Papers
- Pucel, 2002-03 Survey Reports
- Mark, Overall Evaluation

Dissemination
- Develop and maintain access to ATE materials on WMU website
- Focus on creating products of interest to the field
- Adapt standard reports and papers to be better suited to audience:

Evaluation Outputs

Survey Results
- Annual Reports 2001-2004
- Fact Sheets 2005 (6)

Site Visit Reports (13)
Provided to sites only; not published; not given to NSF

Issue Papers (9): In place of planned cross-case study
- Collaboration
- Dissemination
- Materials Development
- Professional Development,
- Program improvement
- Recruitment and retention
- Sustainability
- Advisory Committees
- Evaluation

Other Reports (total ~40)
- Report on Issues 1-7 (above)
- Report on Issues 8-9 (above)
- Presentations at Conferences

Targeted Studies (4)
- Value-added to business/industry
- Materials development
- Professional development
- Sustainability

Special Products
- Brochure versions of targeted studies
- Web-based access survey results
- Handbook for Natl Visiting Committee
- Guide for conducting site visits
- Publications in academic journals

Eval Outcomes /Impact

NSF Reports to Congress based on evidence of quality and accomplishments of ATE projects and centers.

Mid-course program adjustments based on data result in increased performance and effectiveness.

Knowledge production

Better informed STEM community due to more user-friendly documents:
- Papers synthesizing site visits and survey data
- Brochures
- Handbooks

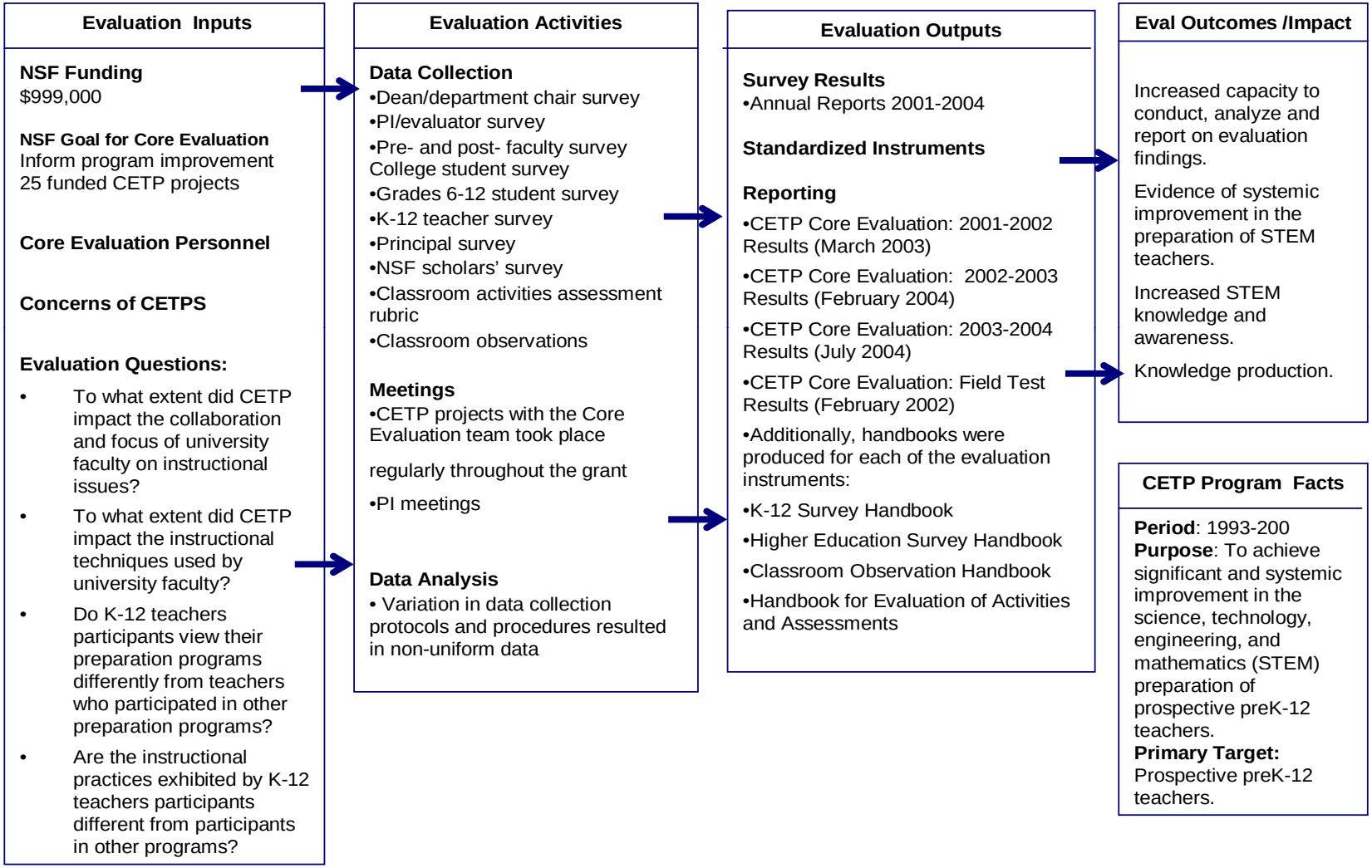
Increased STEM community awareness.

More **frequently used** evaluation results.

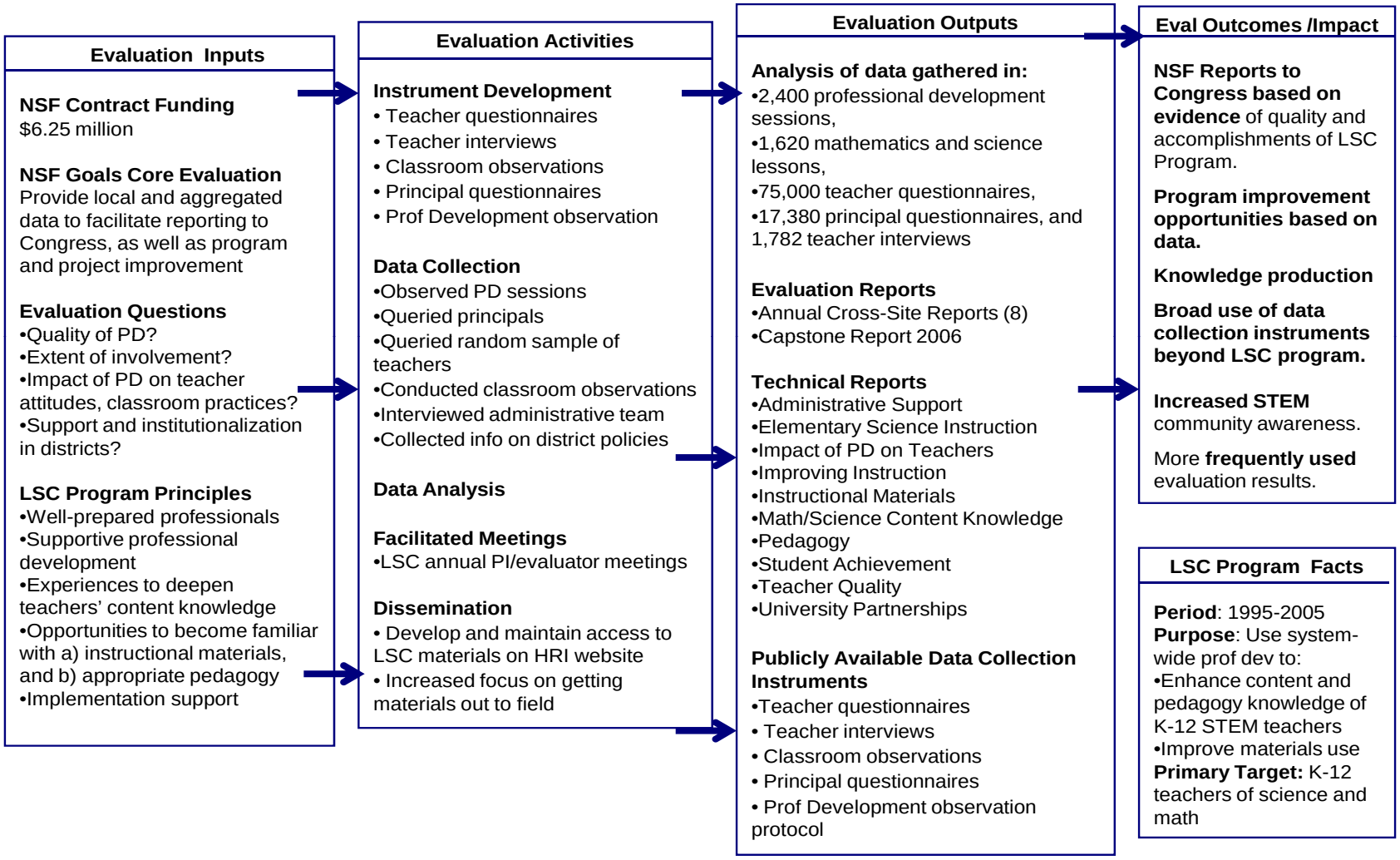
ATE Program Facts

Period: 1993-2005
Purpose: (1) Build capacity to provide advance tech education; (2) Increase number of skilled technicians.
Primary Target: students & teachers in 2-year colleges and collaborating institutions

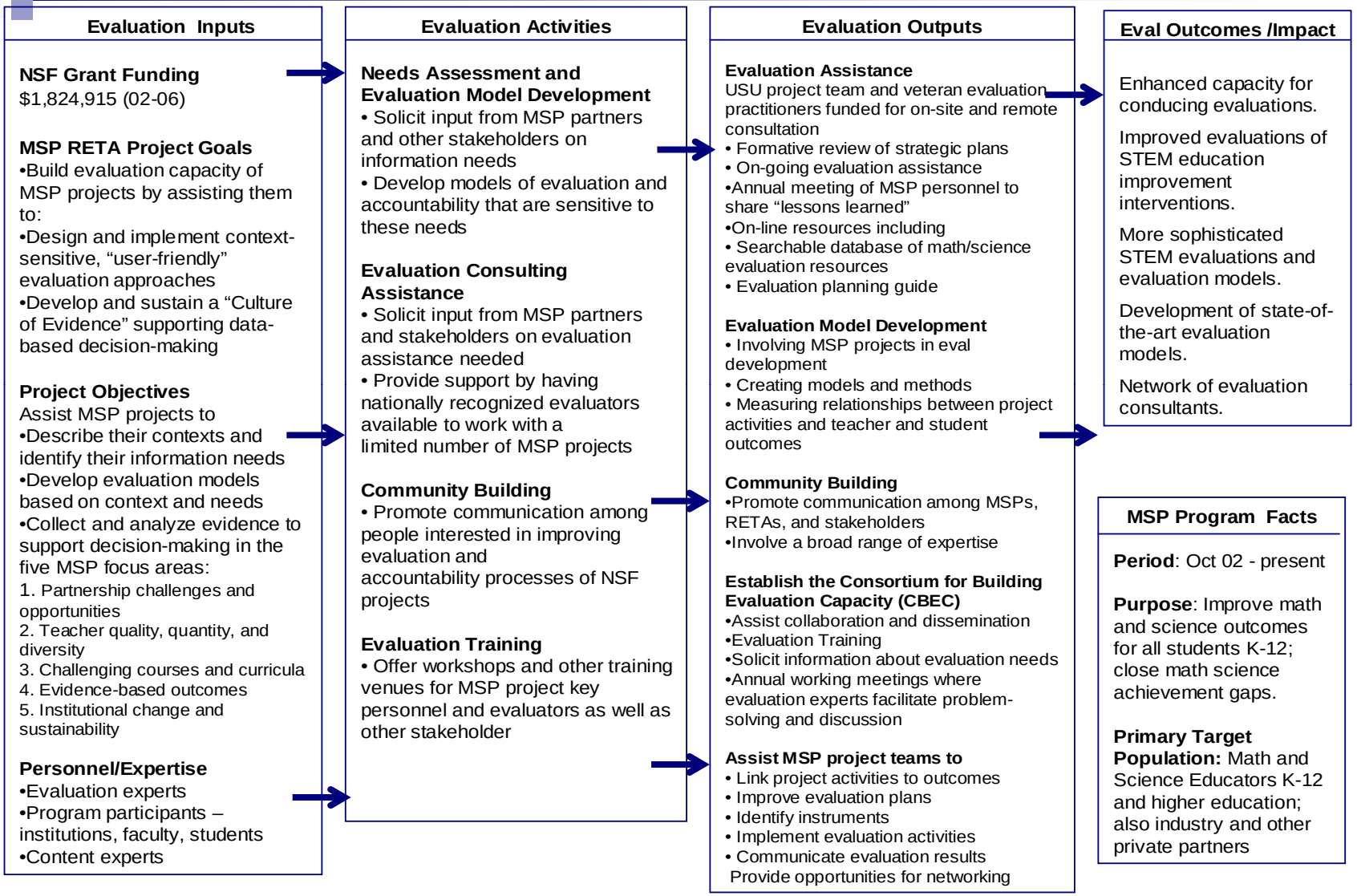
CETP Core Evaluation Logic Model [1/25/2007]



Local Systemic Change through Teacher Enhancement Core Evaluation Logic Model



Utah State MSP-RETA Logic Model [1/31/2007]



Cross-case Findings/Themes

THEME	ATE	CETP	LSC	MSP
Lifecycles: Program-evaluation level	• The evaluation team invited input from different groups at different times.	• The CETP Core Evaluation was implemented after many projects had already been funded and had their own project-level evaluations in place.	• The Program evaluation was new, so some projects were reluctant to participate due to unfamiliarity. • Timing of funding affects use.	• There were differences in timing—the RETA was funded initially with a few MSPs but then other MSPs were added.
Lifecycles: Project-evaluation level	• Project perceptions of the annual survey changed during the project's duration.	• "Mature" CETPs and "developing" CETPs had different experiences.	• Increased conflict between the projects and the Core when the project evaluations pre-date core evaluation. • More impact on field after program done.	• Both the MSPs and the RETA went through significant evolution. • Need for working from the start to the end of MSPs' evaluation projects.
Lifecycles: Individual level		• Previous evaluation experience. • Other individual factors (retirements, moves, etc.).	• Experience level of local evaluations affects attitude toward Core. • Taking the long view on give-and-take affects attitude toward involvement.	• Previous evaluation experience.
Project control	• Project-Level Control: The evaluation team controlled the evaluation. • Program-Level Control: NSF offered input, not instruction.	• Participation in all CETP core evaluation activities was strictly voluntary.	• Funding mechanism prescribed level of control. • Involvement fostered ownership.	• No intended control/requirements to participate. • Emphasis on rigorous evaluation • Some projects felt that they were required to participate.

Cross-case Findings/Themes (Cont.)

THEME	ATE	CETP	LSC	MSP
Project uniqueness/ Alignment between the program and project evaluations	• It was difficult to please all the projects, all the time; but some projects were pleased, some of the time.	• Considerable effort was made to get input from the projects. • Still, several projects felt that the program evaluation did not fit their context.	• Projects viewed Core as not aligned with projects. • Mandatory participation in the Core evaluation facilitated more sophisticated data analysis. • Some instruments used as vehicles for change. • NSF perceived that their participation in planning enhanced alignment. • NSF perceived that collaborative development of Core established trust.	• Intention to meet local contextual evaluation needs. • The Utah State MSP-RETA's framework was not applicable to some MSP projects.
Other	• Collaboration: Projects appreciated the opportunities for interaction. • Dissemination: The evaluation had increased impact because of its dissemination as well as the framework for projects to do their own dissemination.	• Issues preventing extensive dissemination. • Building a network of projects.	• Participation contributed to the development of a sense of community and builds trust. • Data collection served a variety of roles. • Different people in the same program had different perceptions of involvement. • Communication played important role in participants' perceptions of an evaluation.	• Responsiveness and expertise of the MSP-RETA leadership. • Networking opportunities: Participation in eval conferences. • Rigorous approach: Use of rigorous evaluation models and analysis. • Dissemination: Writing and presenting evaluation reports.